**Step 4 : Test statistic**

Under null hypothesis $H_0$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

follows chi-square distribution with ($n$-1) degrees of freedom.

**Step 5 : Calculation of test statistic**

The value of chi-square under $H_0$ is calculated as

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{8 \times 5.4}{9} = 4.8$$

**Step 6 : Critical value**

Since $H_1$ is a one-sided alternative, the critical values at $\alpha$ =0.05 is $\chi_e^2 = \chi^2_{8,\,0.05} = 15.507$.

**Step 7 : Decision**

Since it is a one-tailed test, the elements of the critical region are determined by the rejection rule $\chi_0^2 > \chi_e^2$.

For the given sample information, the rejection rule does not hold , since $\chi_0^2 = 4.8 < \chi^2_{8,\,0.05} = 15.507$. Hence, $H_0$ is not rejected in favour of $H_1$. Thus, the population variance can be regarded as 9.

## Example 2.8

A normal population has mean $\mu$ (unknown) and variance 0.018. A random sample of size 20 observations has been taken and its variance is found to be 0.024. Test the null hypothesis $H_0$: $\sigma^2 = 0.018$ against $H_1$: $\sigma^2 < 0.018$ at 5% level of significance.

***Solution:***

**Step 1 : Null Hypothesis** $H_0$: $\sigma^2 = 0.018$.

*i.e.* Population variance regarded as 0.018.

**Alternative hypothesis** $H_1$: $\sigma^2 < 0.018$.

*i.e.* Population variance is regarded as lessthan 0.018.

**Step 2 : Data**

Sample size ($n$) = 20

Sample variance ($s^2$) = 0.024

**Step 3 : Level of significance**

$\alpha$ = 5%

**Step 4 : Test statistic**

Under null hypothesis $H_0$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

follows chi-square distribution with ($n$–1) degrees of freedom.



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